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SUBJECT CODE NO: - RNP-8002
FACULTY OF SCIENCE AND TECHNOLOGY
F.E. (Group A & B) Mech/ Civil / CSE / AI & ML / EEE (NEP) Sem - II
Examination June/July 2025
Applied Mathematics-II

[Time: 3 Hours]

[Max. Marks:60]

N. B

Please check whether you have got the right question paper.

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve **any four** questions from Q. nos 2, 3, 4, 5, 6 and 7

SECTION A

Q1 Solve the following questions (Any Ten)**20**

- a. The mode of the dataset {4, 6, 6, 8, 9, 6, 10} is:
 - A) 4 B) 6 C) 8 D) 10
- b. Quartiles divide a dataset into how many equal parts?
 - A) 2 B) 3 C) 4 D) 5
- c. If all the values in a dataset are equal, then the quartile deviation is:
 - A) Maximum B) Minimum
 - C) Zero D) Infinity
- d. If the range of a dataset is 50 and the minimum value is 20, what is the maximum value?
 - A) 50 B) 70 C) 30 D) 100
- e. If mean = 40, median = 38, and standard deviation = 5, Karl Pearson's coefficient of skewness is: _____
 - A) 1.2 B) 0.8 C) 0.4 D) 0.2
- f. The curve $y = x^3$ is symmetrical about
 - A) x-Axis B) y-Axis
 - B) both axis D) Opposite quadrant
- g. The length of the cardioid is $r = a(1 + \sin \theta)$
 - A) 8a B) $8a\sqrt{2}$ C) $a\sqrt{2}$ D) $4a\sqrt{2}$
- h. The value of $\int_0^{\frac{\pi}{2}} \sin^6 x dx$ is
 - A) $\frac{5\pi}{32}$ B) $\frac{\pi}{32}$ C) $\frac{5}{32}$ D) 1

- i. The value of $\int_0^{\infty} e^{-x} x^{-1/2} dx$ is _____
 A) 1 B) $\sqrt{\pi}$ C) 0 D) $\frac{1}{2}$
- j. The value of $\int_0^1 \int_0^{1-x} (x+y) dx dy$ is _____
 A) 2 B) $\frac{1}{3}$ C) $\frac{1}{2}$ D) 1
- k. The double integral $\iint f(x,y) dx dy$ becomes in polar form.
 A) $\iint f(r,\theta) dr d\theta$ B) $\iint f(r,\theta) r dr d\theta$
 C) $\iint f(r,\theta) \theta dr d\theta$ D) $\iint f(x,y) dx dy$
- l. The region enclosed by $x^2 + y^2 = 1$ is _____.
 A) Square B) Circle
 C) Ellipse D) Triangle
- m. The area bounded by the curve $y = x$ & $y = x^2$ is given by _____.
 A) $\int_0^1 (x - x^2) dx$ B) $\int_0^1 (x^2 - x) dx$
 C) $\int_0^1 x^2 dx$ D) $\int_0^1 x dx$

SECTION B

- Q2** a) Find mean mode & median for the following data **05**

C.I	0-10	10-20	20-30	30-40	40-50
f	3	5	8	3	1

- b) Calculate standard deviation **05**

C.I	0-10	10-20	20-30	30-40	40-50	50-60	60-70
f	6	5	8	15	7	6	3

- Q3** a) Trace the curve $a^2 y^2 = x^2 (a^2 - x^2)$ with justification **05**

- b) Trace the curve $r = a(1 + \cos \theta)$ with justification **05**

- Q4** a) Evaluate $\int_0^{\log 2} \int_0^x \int_0^{x+y} e^{x+y+z} dx dy dz$ **05**

- b) Evaluate: $\iint r \sin \theta dr d\theta$ over the cardioid $r = a(1 - \cos \theta)$ above the initial line **05**

Q5 a) Calculate Bowleys coefficient of skewness.

05

C.I	0-10	10-20	20-30	30-40	40-50	50-60	60-70
f	6	5	8	15	7	6	3

b) Find the variance and coefficient of variance

05

Age (in yrs)	10-19	20-29	30-39	40-49	50-59	60-69	70-79
Frequency	03	61	223	137	53	19	04

Q6 a) Evaluate $\int_0^{2a} x\sqrt{2a-x} \, dx$

05

b) Prove that $\int_0^{\infty} \frac{dx}{1+x^4} = \frac{\pi\sqrt{2}}{4}$

05

Q7 a) A quadrant of a circle $x^2 + y^2 = a^2$ revolves round its chord. Find the Surface area of the solid thus formed.

05

b) Find the Volume bounded by the cylinder $x^2 + y^2 = 4$ and planes $y + z = 4$ & $z = 0$

05